

Chapter 11 Angular Momentum

PHYS 1408

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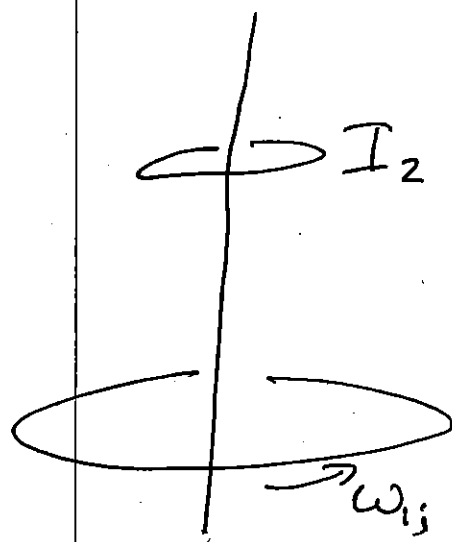
$$\vec{L} = I \vec{\omega} \quad \text{1-D} \quad L = I\omega$$

Conservation of Angular Momentum $L_i = L_f$

inelastic collisions

Drop a disk on another rotating disk

$$I_1 = \frac{1}{2}MR^2 \quad M=500\text{kg} \quad R=2\text{m} \quad \omega_{1i} = 2 \text{ rad/s}$$



$$I_2 = \frac{1}{2}mr^2 \quad m=200\text{kg} \quad r=1\text{m}$$

$$\omega_{2i} = 0$$

2 Drops on Disk 1.

It hits 1, bounces, chatters, and the two stick.

$$\omega_f = 0$$

$$L_i = L_f \quad I_i \omega_i = I_f \omega_f$$

$$\omega_f = \frac{I_i}{I_f} \omega_i = \frac{\frac{1}{2}MR^2 \omega_i}{\frac{1}{2}MR^2 + \frac{1}{2}mr^2}$$

$$\omega_f = \frac{\frac{1}{2} 500 2^2 2 \text{ rad/s}}{\frac{1}{2} 500 2^2 + \frac{1}{2} 200 1^2} = \frac{2000}{1100} = 1.8 \text{ rad/s}$$

Example 2

Merry go round

$$I = \frac{1}{2} MR^2$$

$$M = 500 \text{ kg}$$

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$$R = 2 \text{ m} \quad \omega_i = 1 \frac{\text{rad}}{\text{s}}$$

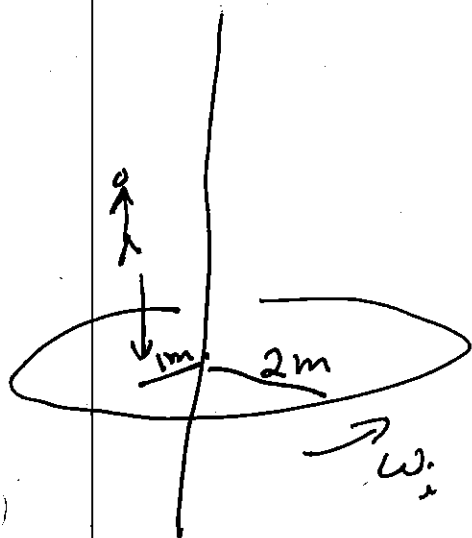
and a 100 kg man drops onto the disk
at $r = 1 \text{ m}$, $\omega_f = ?$

$$L_i = L_f \quad \frac{1}{2} MR^2 \omega_i = \left(\frac{1}{2} MR^2 + I_{\text{man}} \right) \omega_f$$

I_{man} by // axis theorem

$$I_m = I_o + m d^2$$

$$I_o = 0 \quad m = 100 \text{ kg} \quad d = 1 \text{ m}$$



$$\omega_f = \frac{\frac{1}{2} MR^2 \omega_i}{\frac{1}{2} MR^2 + m r^2}$$

$$\omega_f = \frac{\frac{1}{2} 500 2^2 1 \frac{\text{rad}}{\text{s}}}{\frac{1}{2} 500 2^2 + 100 1^2} = \frac{1000}{1100} = 0.91 \frac{\text{rad}}{\text{s}}$$

Rotating disk $I = \frac{1}{2} MR^2$ $M = 500 \text{ kg}$

$$R = 2 \text{ m} \quad \omega_{1i} = 3 \text{ rad/s}$$

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has a counter-rotating ring fall on it.

$$i = mr^2 \quad r = 1.5 \text{ m} \quad m = 200 \text{ kg} \quad \omega_{2i} = -2 \frac{\text{rad}}{\text{s}}$$

$$L_i = L_f \quad I\omega_{1i} + i\omega_{2i} = (I + i)\omega_f$$

$$\omega_f = \frac{\frac{1}{2} MR^2 \omega_{1i} + mr^2 \omega_{2i}}{\frac{1}{2} MR^2 + mr^2}$$

$$\omega_f = \frac{\frac{1}{2} 500 2^2 3 \frac{\text{rad}}{\text{s}} + 200 1.5^2 (-2 \frac{\text{rad}}{\text{s}})}{\frac{1}{2} 500 2^2 + 200 1.5^2}$$

$$\omega_f = \frac{3000 + -900}{1000 + 450}$$

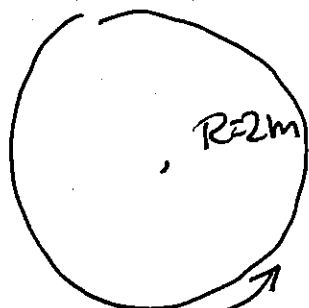
$$\omega_f = \frac{2100}{1450} = 1.45 \frac{\text{rad}}{\text{s}}$$

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100 kg man runs up to turning
MGR, Jumps on rim.

$$I = \frac{1}{2} MR^2$$

$$M = 500 \text{ kg}$$



men $\rightarrow$
 $m = 100 \text{ kg}$
 $v = 3 \text{ m/s}$

$$\omega_i = 0.5 \frac{\text{rad}}{\text{s}}$$

$$L_i = L_f$$

$$\frac{1}{2} MR^2 \omega_i + mvR = \left(\frac{1}{2} MR^2 + mR^2 \right) \omega_f$$

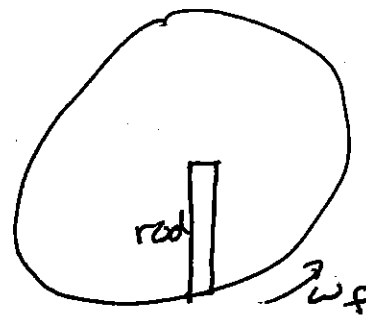
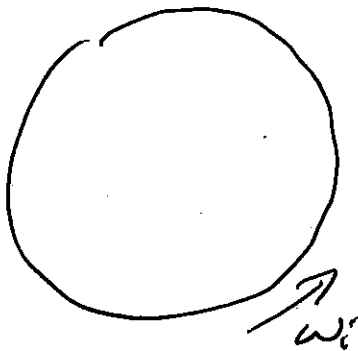
$$\omega_f = \frac{\frac{1}{2} MR^2 \omega_i + mvR}{\frac{1}{2} MR^2 + mR^2}$$

$$\omega_f = \frac{\frac{1}{2} 500 2^2 0.5 \frac{\text{rad}}{\text{s}} + 100 3 \frac{\text{m}}{\text{s}} 2}{\frac{1}{2} 500 2^2 + 100 2^2}$$

$$\omega_f = \frac{500 + 600}{1000 + 400} = \frac{1100}{1400} = 0.786 \frac{\text{rad}}{\text{s}}$$

500 kg 2m DISK MGR at $2 \frac{\text{rad}}{\text{s}}$
 and we drop a 2m rod $m=100\text{kg}$
 from center to edge

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$$I_{\text{rod}} = \frac{1}{3} m l^2$$

$$\omega_f = ?$$

$$L_i = L_f$$

$$\frac{1}{2} M R^2 \omega_i = \left(\frac{1}{2} M R^2 + \frac{1}{3} m l^2 \right) \omega_f$$

$$\omega_f = \frac{\frac{1}{2} M R^2 \omega_i}{\frac{1}{2} M R^2 + \frac{1}{3} m l^2} = \frac{\frac{1}{2} 500 \cdot 2^2 \cdot 2 \frac{\text{rad}}{\text{s}}}{\frac{1}{2} 500 \cdot 2^2 + \frac{1}{3} 100 \cdot 2^2}$$

$$\omega_f = \frac{2000}{1000 + \frac{1}{3} 400} = \frac{2000}{1133} = 1.77 \frac{\text{rad}}{\text{s}}$$

$$\omega_i = 50 \frac{\text{rad}}{\text{s}}$$

$$\omega_f = 100 \frac{\text{rad}}{\text{s}}$$

$$t = 10 \text{ s}$$

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$$\alpha = ?$$

$$\omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i) \quad \text{NO}$$

$$\omega_f = \omega_i + \alpha t$$

$$\alpha = \frac{\omega_f - \omega_i}{t}$$

$$\alpha = \frac{100 - 50}{10} = \frac{50}{10} = 5 \frac{\text{rad}}{\text{s}^2}$$

or $\omega_i = 50 \frac{\text{rad}}{\text{s}} \quad \omega_f = 100 \frac{\text{rad}}{\text{s}} \quad \text{and} \quad \Delta\theta = 40 \text{ rad}$

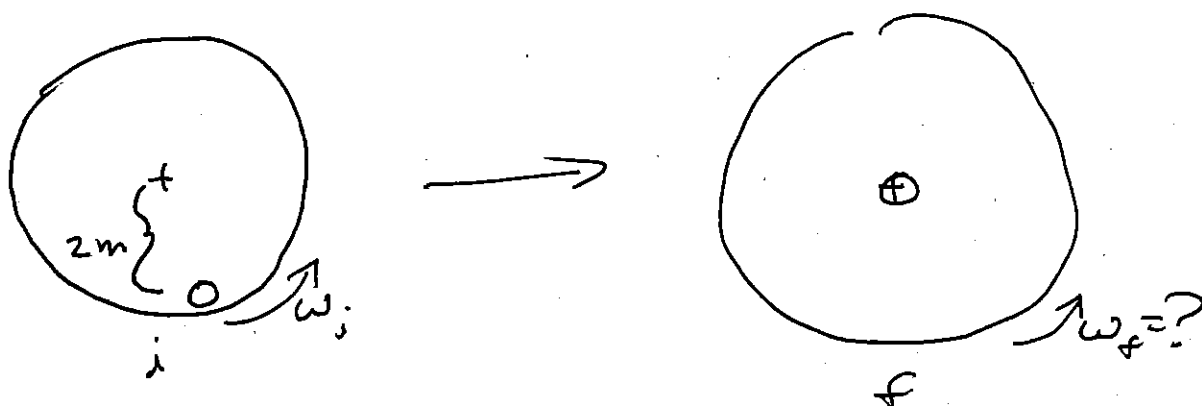
$$\omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i)$$

$$\alpha = \frac{\omega_f^2 - \omega_i^2}{2(\theta_f - \theta_i)} = \frac{100^2 - 50^2}{2 \cdot 40}$$

$$\alpha = \frac{10^4 - 2.5 \cdot 10^3}{80} = \frac{7.5 \cdot 10^3}{80} = 93.75 \frac{\text{rad}}{\text{s}^2}$$

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A 100 kg man rides on the rim of a MGR
 $I = \frac{1}{2} MR^2$ $M = 500 \text{ kg}$ $R = 2 \text{ m}$. He walks from
 the rim to the center. Initially the angular
 velocity is 2 rad/s . What is the angular
 velocity when he reaches the center?



It's a conservation of angular momentum
 problem $L_i = L_f$

$$L_i = I_i \omega_i = \left(\frac{1}{2} MR^2 + md^2 \right) \omega_i$$

M is the MGR
 m is the man
 d is the distance
 from the axis $2m$

$$L_f = I_f \omega_f = \left(\frac{1}{2} MR^2 + md^2 \right) \omega_f$$

d is zero /
 he reaches the axis

$$\omega_f = \frac{\frac{1}{2} MR^2 + mR^2}{\frac{1}{2} MR^2} \omega_i = \frac{\frac{M}{2} + m}{\frac{M}{2}} \omega_i = \frac{M+2m}{M} \omega_i$$

$$\omega_f = \frac{500 + 2 \cdot 100}{500} 2 \frac{\text{rad}}{\text{s}} = \frac{7}{5} 2 = \boxed{2.8 \frac{\text{rad}}{\text{s}}}$$

How does the Energy change?

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$$E_i = \frac{1}{2} I_i \omega_i^2 = \frac{1}{2} \left(\frac{1}{2} MR^2 + mR^2 \right) \omega_i^2$$

$$= \frac{1}{2} \left(\frac{500}{2} 2^2 + 100 2^2 \right) 2^2$$

$$= 2 (1000 + 400) = 2800 \text{ J}$$

$$E_f = \frac{1}{2} I_f \omega_f^2 = \frac{1}{2} \frac{1}{2} MR^2 \omega_f^2$$

$$= \frac{1}{4} 500 2^2 2.8^2 = 500 2.8^2$$

$$= 3920 \text{ J}$$

It took energy to move from rim to the center.